

Good evening everyone



class start at 8:10

OSI model

IP address

Jatin, H.NO-142
Pincode - 202010

slipknot

warehouse



Lucknow

pincode $\rightarrow$ IP address

it is used to reach your LAN.

it operates at network layer

mac address

it is also known as physical address or hardware address



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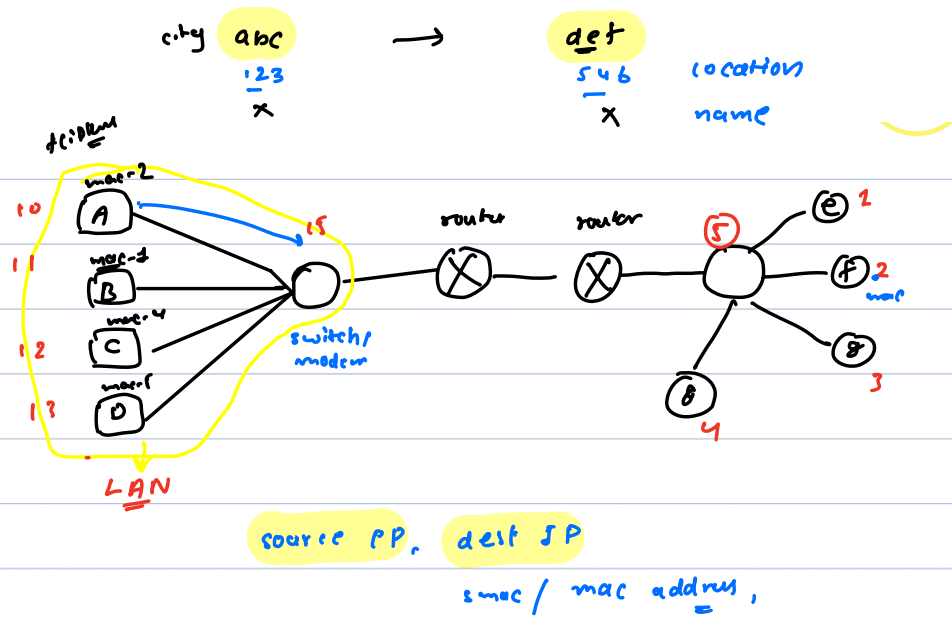
slipknot

warehouse



Lucknow

H.NO $\rightarrow$ mac address



it operates on data link layer.

it represented as six pairs of hexadecimal digit

48 bits

total combination 2^{48}

1A - 2B - C3 - 9F - 22 - 5A



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slipknot

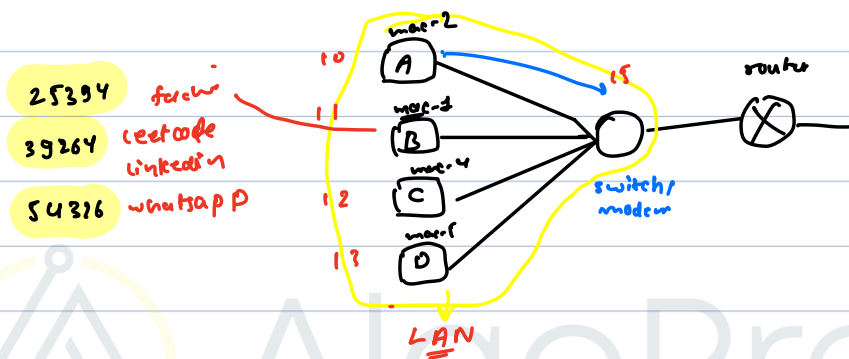
warehouse



LUCKNOW

Port number

it is the part of addressing information
it is used to deliver data packets to
the right/correct application.



Port number → right person

range of port number

(0 - 65535)

fixed port numbers : →
http → (80)
https → (443)
smtp → (25) email

dynamic port : →

chrome → (52353)

it is assigned by operating system.

source IP, dest IP
src mac / mac address,
src port, dest port

break till 9 : 15

DNS (domain name system)

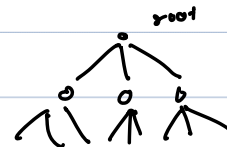
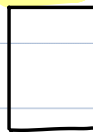
these domain name is easy to remember

ip address		dns
google.com	→ 128.145.32.9	
facebook.com	78.135.2.9	

it is provided by ISP

google.com

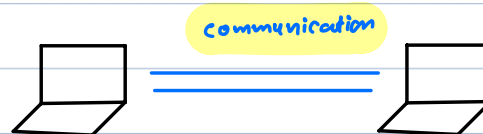
cache



its stored in hierarchical form

Application layer

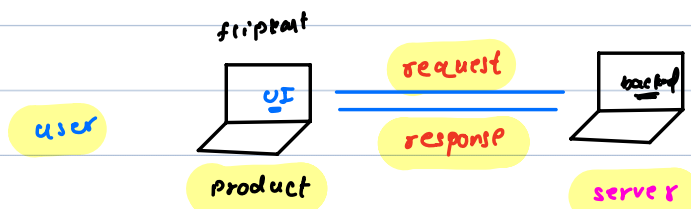
generate data to be sent



1) client server architecture

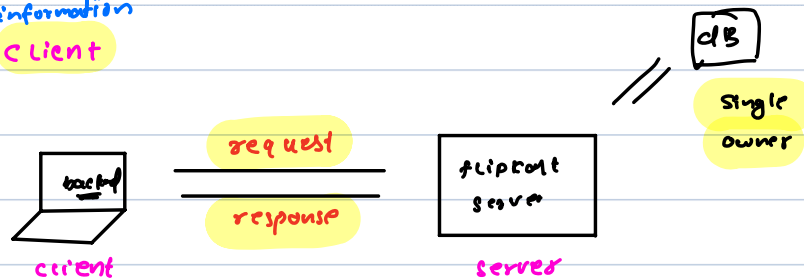
2) Peer to peer architecture

whatsapp / facebook / instagram



get the information

client

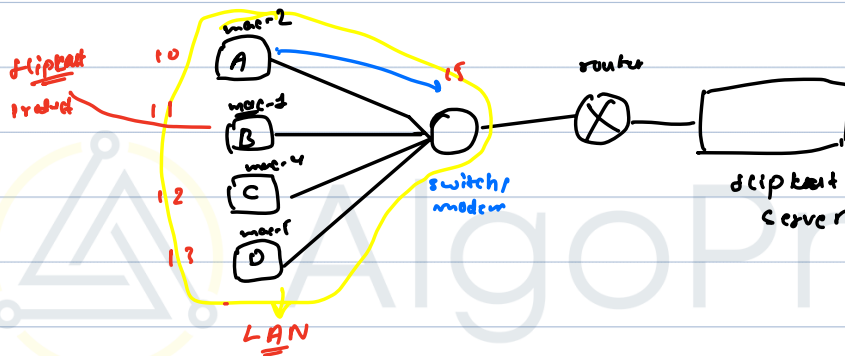


(request response)

client doesn't have any data

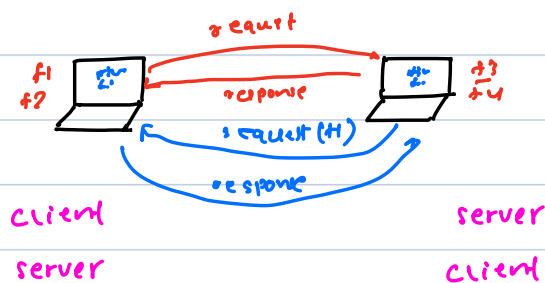
server owns the data

centralised system



2) peer to peer architecture (P2P)

In this any device act as client as well as server.



there is no external database

decentralised system

peer to peer (Torrent) / block chain /
skype

peers $\begin{cases} \text{download} \\ \text{upload} \end{cases}$

seeder $\rightarrow$ to give data
leechers $\rightarrow$ take only

downloading / uploading
data data
receive sent

